

D BHARGAVA RAMA SARMA

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Education

PES University • Aug 2023 – May 2027

Bachelor of Technology in Artificial Intelligence and Machine Learning • Bengaluru, Karnataka

Work Experience

Cyient • Jun 2026 - Jul 2026

Software Engineering Intern • Bengaluru, Karnataka

- Developed an OCR tool, building a document understanding pipeline that combines CNN-based layout detection, transformer-based text recognition, and attention mechanisms to extract structured fields from scanned documents with 88% extraction accuracy. Designed the document quality analysis and preprocessing layer, implementing image denoising, deskewing, and adaptive thresholding across the ingestion pipeline to improve reliability on low-quality scanned inputs.

Centre for Healthcare Engineering and Learning • Jun 2025 - Jul 2025

Research Intern • Bengaluru, Karnataka

- Trained XGBoost and k-NN classification models using librosa audio features and CNN image features to detect Parkinson's disease from voice recordings and handwriting-based wave and spiral patterns across three input modalities, achieving 85% classification accuracy. Engineered a Flask web application enabling real-time sample upload and AI-powered symptom analysis across all 3 input modes, integrating all trained models behind a single interface.

Projects

Real-Time Collaborative Drawing Board with Mini-RAFT | *Node.js, Express.js, WebSocket, Docker, Docker Compose*

- Implemented a RAFT-inspired consensus algorithm across a 3-node cluster with leader election, log replication, and state recovery after restarts. Built a collaborative canvas where drawing operations are replicated across nodes and broadcast to clients via WebSocket, orchestrated with Docker Compose.

Global Conflict and Event Tracker | *Next.js, TypeScript, Supabase, Perplexity AI, Leaflet*

- Architected a real-time intelligence dashboard using Perplexity AI for NLP-based event extraction and severity classification, with hourly ingestion. Built an interactive geospatial map with React-Leaflet and Tailwind CSS, backed by a Supabase PostgreSQL database for event storage.

Privacy-First KYC Verification with Zero-Knowledge Proofs | *Circom, snarkjs, Solidity, Node.js, Express.js, Tesseract.js*

- Built a KYC system that OCRs an ID document, hashes the extracted fields, and generates Groth16 zero-knowledge proofs (age, name, gender) without storing or revealing raw identity data. Deployed Solidity verifier contracts to the Polygon Amoy testnet, enabling gas-free on-chain re-verification of proofs generated with snarkjs.

Skills

Languages: Python, JavaScript, TypeScript, Java, C++

Web & Backend: React, Next.js, Node.js, Express.js, FastAPI, Flask

Databases: MongoDB, PostgreSQL, Supabase

ML/AI & Tools: TensorFlow, Keras, XGBoost, Scikit-learn, OpenCV, Pandas, NumPy, Docker, Git

Certifications

- **Machine Learning Specialization** (Supervised ML, Advanced Learning Algorithms) – DeepLearning.AI, Coursera
- **Licensed Amateur Radio Operator**, Callsign VU3FHA – Department of Telecommunications, Government of India
- **Blockchain Basics** – Cyfrin Updraft

Achievements & Extracurricular Activities

- **State Champion & Zonal Runner-up** at the RBI90 National Quiz Competition (Nov-Dec 2024); also placed top 3 at multiple inter-college and inter-university open quizzes.
- **Marketing and Operations Head, QQC Quiz Club, PES University** (Dec 2023 - May 2026): organized college and inter-college quizzes, curating question sets across science, technology, and current affairs.
- **Marketing Head, MUNSoc, PES University** (Sep 2024 - May 2026): delegated at multiple MUN conferences and managed logistics for UniCon, an inter-college Model UN conference.
- **Volunteer, Akshaya Patra Foundation** (Aug 2023): contributed to community meal preparation initiatives serving underprivileged children.